

Bost-Connes $p = 7$ $(\mathbb{Z}/7\mathbb{Z})^*$, $|G| = 6$ (论文④: 12 fermion 质量谱)

CRT $(\mathbb{Z}/7\mathbb{Z})^* \cong \mathbb{Z}/2 \times \mathbb{Z}/3$

$\mathbb{Z}/2\mathbb{Z}$
 T_3 chirality

$\mathbb{Z}/3\mathbb{Z}$
色 SU(3)

6 个 character $|\tau| = \sqrt{7}$

χ_0 trivial
order 1, 实

$\mathbb{Z}/2$: triv
 $\mathbb{Z}/3$: triv

$L = \zeta$
(真空/起点)

χ_3
order 2, 实

$\mathbb{Z}/2$: 非平凡
 $\mathbb{Z}/3$: triv

$L_3 = 1.147$
(实, self-pair)

χ_2
order 3, 复

$\mathbb{Z}/2$: triv
 $\mathbb{Z}/3$: 非平凡

$|L_2| = 0.318$
 $|L_2|^2 = 0.048$

$\chi_4 = \bar{\chi}_2$
order 3, 复

$\mathbb{Z}/2$: triv
 $\mathbb{Z}/3$: 非平凡

$|L_4| = |L_2|$
(conjugate)

χ_1
order 6, 复

$\mathbb{Z}/2$: 非平凡
 $\mathbb{Z}/3$: 非平凡

$|L_1|^2 = 0.347$

$\chi_5 = \bar{\chi}_1$
order 6, 复

$\mathbb{Z}/2$: 非平凡
 $\mathbb{Z}/3$: 非平凡

$|L_5|^2 = 0.616$

12 fermion = 3 generation $\times$ 4 sector (含计算路径 + 节点因子 + Koide K_k)

带电轻子

e, μ, τ ($Q = -1$)
color singlet (实组合)

路径: $\chi_0 \rightarrow \chi_2$

质量公式

$$m = v \cdot |L_2|^4 \cdot \frac{1}{\sqrt{d_1}} \cdot K_k$$

L因子 / 节点 / Gen

$d_1 = 2$ (Galois pair)

K_k = Koide 旋转

$\delta = -12.73^\circ$

质量 (MeV)

	Gen	BC	exp
τ	K_0	1791	1777
μ	K_1	106.5	105.7
e	K_2	0.514	0.511

偏差 < 1%

中微子

$\nu_{e,\mu,\tau}$ ($Q = 0$)
color singlet

路径: $\chi_0 \rightarrow \chi_2 \rightarrow [\chi_0]^2 \rightarrow \chi_2$

χ_3 直接关闭, 借 $\chi_3^2 = \chi_0$ 间接到达

质量公式

$$m = v \cdot |L_2|^8 \cdot \frac{1}{d_1 N^2} \cdot \lambda_k$$

L因子 / 节点 / Gen

$N = \varphi(7^4) = 2058$

λ_k = Hurwitz 权重

(Majorana, no see-saw)

质量 (meV)

ν_3	50.5
ν_2	10.0
ν_1	5.11

m_3 偏差 1%

up 夸克

u, c, t ($Q = +\frac{2}{3}$)
color triplet rgb

路径: $\chi_0 \rightarrow \chi_2 \rightarrow \chi_3 \rightarrow \chi_1$
($\chi_1 = \bar{\chi}_2 \cdot \chi_3$, $|L_1|$ 显式入 L 因子)

质量公式

$$m = v \cdot |L_2|^4 L_3^2 |L_1|^4 \cdot \frac{1}{\sqrt{d_1}} \cdot G$$

L因子 / 节点 / Gen

c: $G = K_0 = 1$

t: $G = 1/\alpha_{EM}$ (Dyson)

u: $G = \lambda^2 \alpha_W$ (Cabibbo)

质量

	Gen	BC	exp
c	K_0	1273	1270
t	$1/\alpha$	174.5G	172.8G
u	$\lambda^2 \alpha_W$	2.12	2.16

偏差 $\leq 2\%$

down 夸克

d, s, b ($Q = -\frac{1}{3}$)
color triplet rgb

路径: $\chi_0 \rightarrow \chi_2 \rightarrow \chi_3 \rightarrow \chi_5$
($\chi_5 = \chi_2 \cdot \chi_3$ char 乘积)

质量公式 (b)

$$m_b = v \cdot |L_2|^4 L_3 \cdot \frac{d_2}{\sqrt{d_1} |\zeta|} \cdot K_0$$

质量公式 (s, d)

$$m = v \cdot |L_2|^4 L_3^2 \cdot \frac{1}{|\zeta|^2} \cdot G$$

$d_2 = 3$ (色解禁)

$|\zeta(\frac{1}{2})| = 1.460$

质量 (MeV)

	Gen	BC	exp
b	K_0	4219	4180
s	K_1	92.9	93.4
d	$K_1 \lambda^2$	4.58	4.67

偏差 $\leq 2\%$

统一质量公式:

$$m_f = v \times \underbrace{\prod_{\text{路径}} |L_k|^{\text{power}}}_{\text{L 因子 (路径)}} \times \underbrace{\prod_{\text{节点}} \text{归一化}}_{\text{节点因子}} \times \underbrace{\text{Gen}_f}_{\text{代间因子}}$$

Koide 旋转: $K_k = (f_k/f_0)^2$, $f_k = 1 + \sqrt{d_1} \cos(\delta + 120^\circ k)$, $\delta = -12.73^\circ$ (Jacobi 相位)

$v = v_{EW} = 246.2$ GeV, $\lambda = 0.2253$ (Cabibbo angle), $\alpha_{EM} = 1/137.04$, $\alpha_W = 1/29.6$